

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO2: Neural Network Representation, Perceptron Model, Multilayer Perceptron's, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)

Assessment Tool Weightage: 10%

QUESTIONS:15

Total Marks:25

Annexure A

DEBUGGING & OPTIMIZATION TASK QUESTIONS

Code of Conduct:

I Savitha R / 192421368 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: SAVITHA R

1. Perceptron fails to converge

Conceptual Understanding

A perceptron is designed to classify **linearly separable data** and normally converges after a finite number of updates. Failure to converge may indicate problems with the learning parameters or implementation.

Accuracy

Possible causes include:

- Improper **learning rate**
- Incorrect **weight initialization**
- Bias not included or updated correctly
- Incorrect perceptron **weight update rule**

Application of Knowledge

To debug, check the weights, bias, prediction, and error after every iteration using a small dataset. Verify the update rule:

$$\mathbf{w}(\text{new}) = \mathbf{w}(\text{old}) + \eta(\mathbf{y} - \hat{\mathbf{y}})\mathbf{x}$$

Choose a suitable learning rate and normalize the input features. Ensure that the bias is updated correctly.

Completeness

Use multiple iterations and monitor the number of classification errors. Stop when there are no errors.

Clarity & Presentation

Thus, correct implementation, suitable learning rate, proper initialization, feature normalization, and correct bias handling can provide **faster and stable convergence**.

2. MLP using Backpropagation is slow

Conceptual Understanding

Backpropagation adjusts network weights using gradients to minimize the error. Slow convergence can occur because of **vanishing gradients, unsuitable activation functions, poor initialization, or inappropriate learning rates**.

Accuracy

Common problems are:

- Vanishing gradients
- Poor weight initialization
- Very high or low learning rate
- Unnormalized input data
- Unsuitable activation functions

Application of Knowledge

First, monitor training and validation loss. Then check the gradient values in each layer. Use **ReLU** for hidden layers where suitable and Xavier/He initialization. Normalize the input data and select an appropriate learning rate.

Advanced optimizers such as **Adam or RMSProp** can also improve convergence.

Completeness

Batch normalization and learning-rate scheduling can further improve training stability.

Clarity & Presentation

Therefore, proper activation functions, initialization, normalization, and adaptive optimizers help the MLP **converge faster and avoid training difficulties**.

3. Genetic Algorithm gives inconsistent solutions

Conceptual Understanding

A Genetic Algorithm searches for good solutions using **selection, crossover, and mutation**.

Different results across runs may occur because of randomness, poor chromosome representation, incorrect fitness evaluation, or loss of population diversity.

Accuracy

Possible causes include:

- Improper chromosome encoding
- Incorrect fitness function
- Premature convergence
- Very low mutation rate
- Excessive selection pressure

Application of Knowledge

First, verify that the chromosome correctly represents the solution. Test the fitness function using known solutions. Monitor population diversity and fitness across generations.

If diversity decreases too quickly, increase the mutation rate or reduce selection pressure. Use **elitism** to preserve the best individuals.

Completeness

Run the algorithm with different random seeds and compare the final fitness values.

Clarity & Presentation

Hence, correct encoding, a suitable fitness function, balanced mutation and selection, and elitism can improve **solution quality and convergence**.

5. Decision Tree overfits

Conceptual Understanding

A decision tree may achieve very high training accuracy but poor test accuracy because of **overfitting**. A very deep tree can memorize training data instead of learning general patterns.

Accuracy

Possible causes include:

- Excessive tree depth
- Too many branches
- Noisy data
- Irrelevant features
- Insufficient pruning

Application of Knowledge

Compare training and validation accuracy to identify overfitting. Use **cross-validation** to evaluate the model. Limit the maximum tree depth and minimum samples per leaf.

Apply **pre-pruning or post-pruning** to remove unnecessary branches. Feature selection can also reduce noise.

Completeness

Testing the pruned tree on unseen data helps verify whether generalization has improved.

Clarity & Presentation

Thus, **pruning, cross-validation, feature selection, and controlling tree depth** can reduce overfitting and improve performance on unseen data.

8. K-Means gives inconsistent clusters

Conceptual Understanding

K-Means assigns data points to the nearest centroid and repeatedly updates the centroids.

Since the initial centroids are important, different initialization can produce different cluster assignments.

Accuracy

Possible reasons are:

- Poor centroid initialization
- Incorrect value of **K**
- Different feature scales
- Presence of outliers

- Non-spherical cluster structure

Application of Knowledge

Normalize or standardize the features before clustering. Use **K-Means++** for better centroid initialization. Select K using the **Elbow method** and evaluate cluster quality using the **Silhouette score**.

Handle extreme outliers where appropriate and run K-Means multiple times.

Completeness

Compare the within-cluster sum of squares and choose the solution with better clustering quality.

Clarity & Presentation

Therefore, **K-Means++**, **normalization**, **proper selection of K**, and **outlier handling** can make the clustering more stable and meaningful.